

# ELI Robot Notes

Version 2 – 5/1/13

**Jonathan Connell**  
[jconnell@us.ibm.com](mailto:jconnell@us.ibm.com)

|                                                  |          |
|--------------------------------------------------|----------|
| <b>Supporting Media</b> (presentations & videos) | <b>2</b> |
| <b>Hardware</b> (switches, charging, etc.)       | <b>3</b> |
| <b>Speech</b> (setting the user)                 | <b>4</b> |
| <b>eli_leash</b> (joystick control)              | <b>5</b> |
| <b>sp_drive</b> (simple voice command)           | <b>6</b> |
| <b>Thursday</b> (person following)               | <b>7</b> |
| <b>Klepto</b> (tabletop manipulation)            | <b>8</b> |
| <b>Troubleshooting</b> (common problems)         | <b>9</b> |

## Supporting Media

**Name** – ELI stands for “Extensible Language Interface”. That is, you control the robot through natural language and it can acquire new nouns and verbs from interaction.

**Project History** – This robot was the result of an IBM Far-Reaching Research (FRR) project during 2011. About 1.5 man-years went into the current robot, including design, fabrication, drivers, and high-level programming.

**Factoids** – The control programs are all written in C++ and the head is an X-Box Kinect.

**Website** – Most of the supporting details (including a paper) can be found on the externally viewable website [http://researcher.watson.ibm.com/researcher/view\\_project.php?id=1403](http://researcher.watson.ibm.com/researcher/view_project.php?id=1403).

**Presentations** – The website has several PowerPoint decks that may be of interest:

***Quick Intro*** – Description of robot capabilities in an Artificial Intelligence context.

***Overview*** – More detailed walkthrough of both the body and arm demo videos.

***Business Case*** (IBM Confidential) – Costs and deployment scenarios for an eldercare robot.

**Videos** – There are various slice-and-dice version of a few basic videos on the website:

***Learning*** (0:54) – Just learning a new object model and the poke procedure.

***Arm Compendium*** (4:35) – Basic manipulation, reasoning, noun and verb learning (4 parts).

***Speech Dialog*** (1:21) – Manipulation and language with pronoun resolution.

***Learning Nouns*** (0:57) – Acquiring and using visual models from speech & gesture.

***Remote Database Access*** (1:21) – Prevention of double dosing, substitution of objects.

***Learning a Verb*** (0:42) – Acquiring the “poke” macro through spoken instruction.

***Body Compendium*** (2:30) – Simple voice command and picking up object on floor (2 parts).

***Simple Speech*** (1:15) – Basic “verbal joystick” shows low-level human-robot interaction.

***Lifting Object*** (1:07) – Joystick control to demonstrate floor pickup in a kitchen.

**Papers** – There is a short paper from the *2012 Artificial General Intelligence (AGI) Conference* available on the web site. There are also several older, somewhat less relevant papers on the value of language for robots and learning.

## Hardware

**USB Cable** – There is a single USB cable that connects the robot to the controlling laptop. Note that the Kinect CANNOT handle USB-3, so avoid blue ports or ones marked “SS”. Generally the yellow port (always powered on) is the safest one to use.

**Power Switches** – For the robot to run, both switches under the speakers must be in the back position (“reset” on the right and “Run” on the left). After this press the “Start” button under the silver microphone bar. The blinking red lights on the sides should come on.

**Emergency Stop** – Slapping either rocker switch near the red blinking light will disable the motors. They will remain off until the switches are flipped back and the “Start” button is pressed.

**Rebooting** – The USB cable needs to be DISCONNECTED to power cycle the robot (else some power is delivered via the USB hub). So unplug the cable, toggle the power switch off then on, then hit the “Start” button and plug the USB cable back into the laptop.

**Charging** – Flip the left switch to “Charge” and the right switch to “Reset” then plug the charger into the connector at the rear of the robot (“Charge”). The charger will turn green when done.

**Short Moves** – Face the robot and use both hands to grab the robot around the neck near the “ELI” label (like Homer Simpson: “Why you ... !”). With this leverage you can tip the robot and push it backwards on the rear blue wheels.

**Remote Operation** – The various programs can be start from a different device (like an iPhone) given a suitable WiFi screen sharing application. UltraVNC works reasonably well and is better than Windows Remote Desktop because it does not try to redirect the audio input and output. For programs that show a video stream, minimize the window to prevent slowdowns due to transfer.

**Velcro Straps** – These are for securing the laptop during motion. The laptop can go open and sideways (if it is small enough) or closed and long ways.

**Removing the Batteries** – To make carrying more manageable, the batteries can be removed to make the robot 40+ pounds lighter. On the right hand side there are two thumbscrews underneath that secure the edge of the batteries. Remove the screws and the side panel then slide the batteries slightly apart. Disconnect the BLACK wires first (to avoid sparking against the ground frame) then the red wires. Now slide both batteries out to the right (the fit is intentionally tight).

**Long Moves** – To transport the robot to another site first lower the lift mechanism so the head is below the top of the pole. Next remove the batteries as described above. Finally, strap the lift part of the elbow the serrated lower rack (with some padding in between). The robot can be pushed around on the rear casters by grabbing the upper serrated rack near the “ELI” label and tipping the robot backwards.

**Tires** – These are air-filled and gradually lose pressure over several months. Reinflate to 36 psi using a normal automotive compressor.

## Speech

**Why Microsoft** – The speech recognition comes standard with Windows 7 (and XP). The robot can be made to work with IBM / Nuance software, but there are no good tools for training new speakers or for altering the language model.

**Setting the User** – In Start / Control Panel select “Speech Recognition” (or use the desktop shortcut). Select “Advanced speech options” then there will be a list box called “Recognition Profiles”. Put a check next to the proper user. The robot programs will extract the name from whatever speech model is set. Alternatively, some programs take this model as an argument.

**Setting the Microphone** – You only need to do this if the robot does not seem to be understanding things. In Start / Control Panel select “Speech Recognition” (or use the desktop shortcut). Select “Advanced speech options” then click the “Audio Input” button. Under the “recording” tab you should set “Voice Tracker II” as the default input device.

**Adjusting the Input Volume** – You generally don’t need to do this, except perhaps when training a new speaker. In Start / Control Panel select “Speech Recognition” (or use the desktop shortcut). Select “Advanced speech options” then click the “Configure Microphone” button. The default “Headset Microphone” setting is fine so click “Next” several times and speak the Peter sentence. Have the computer accept the new input even if it complains about bad recognition.

**Training a New User** – In Start / Control Panel select “Speech Recognition” (or use the desktop shortcut). Select “Advanced speech options” then click the “New” button. Name the profile “FirstName-LastName-VT2” so the robot can parse it. Finally, click the “Train Profile” button. With the robot about 4 feet away and facing you read the various sentences (the wizard automatically advances when it has heard you).

**Noisy Environments** – Microsoft seems to require silence to mark the end of every sentence. If the room is noisy the robot might wait what seems a very long time before acting on what you say. Also, if the audience is laughing because the robot previously made a mistake, it can miss the end silence and thereby mess up the next command as well.

**Bluetooth Headset** – In many cases (especially crowds or at conferences) it makes sense to use a close-talking earbud instead. See “Setting the Microphone” above for how to configure this. The robot will still be able to turn toward the loudest sound, but this might not be the user.

**Robot’s Voice** – Yes, he has a Scottish accent. This is the “Stuart” voice from CereProc. The default Microsoft “Anna” voice can alternatively be selected but it sounds irritating to me. You can try out other voices at <http://www.cereproc.com/en/store> and buy them for \$40 each.

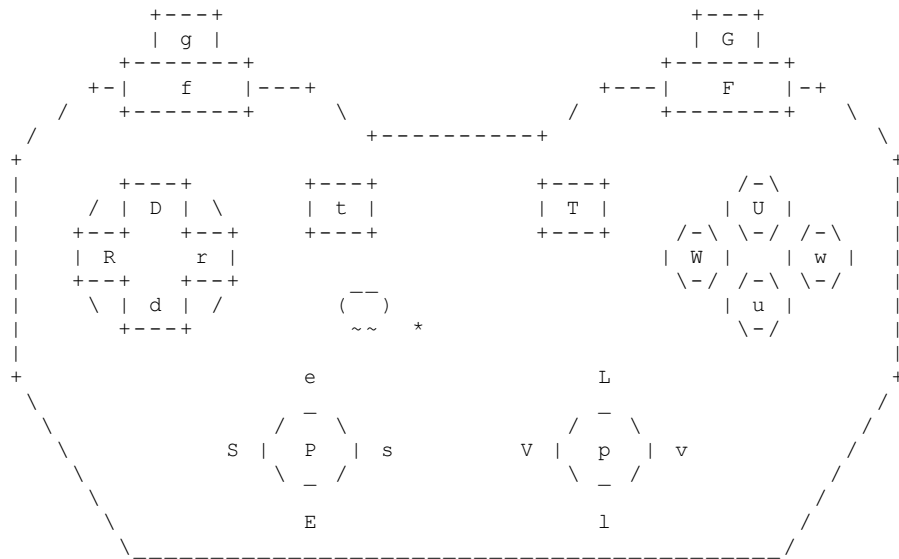
**Changing the Voice** – In Start / Control Panel select “Speech Recognition” (or use the desktop shortcut). Select “Advanced speech options” then click the “Text to Speech” tab. There is a drop down list called “Voice selection” that allows you to choose among installed voices (e.g. Anna).

## eli\_leash

**Gamepad** – This program was designed to be used with a Logitech F-310 Gamepad. Make sure the slide switch on the back is set to “X” (not “D”).

**Command Line** – The program takes no arguments (but ? will give the list of key commands). It will initialize all the robot hardware then show a map of the gamepad controls (below).

**Motor Control** – The program is designed to allow control of each individual motor for debugging. However, this can make it challenging to drive (especially the arm). The various buttons and joysticks ALL have some function as shown below. Note the head pan control is activated by pressing down the center of the joystick until it clicks.



**Keyboard Control** – The various joints can also be controlled by typing single characters on the keyboard. Generally an uppercase letter moves the joint one way and the lowercase version moves it the other.

**F** = raise Forklift stage  
**f** = lower forklift stage  
**D** = Drive forward  
**d** = drive backward  
**R** = Rotate to left  
**r** = rotate to right

**P** = Pan head to left  
**p** = pan head to right  
**T** = Tilt head up  
**t** = tilt head down

**S** = rotate Shoulder to left  
**s** = rotate shoulder to right  
**E** = rotate Elbow to left  
**e** = rotate elbow to right  
**L** = Lift elbow  
**l** = lower elbow

**W** = rotate Wrist counter-clockwise  
**w** = rotate wrist clockwise  
**V** = Veer wrist to left  
**v** = veer wrist to right  
**U** = raise Up fingertips  
**u** = lower fingertips

**G** = close Gripper  
**g** = open gripper

## sp\_drive

**Command Line** – The “sp\_drive” program takes an optional argument (like Jon-Connell-VT2) which is the voice model to use, otherwise it uses the default set in the Control Panel. At startup the robot will be initialized and then announce itself. During operation the command window will show each phrase heard and give its interpretation as a primitive action.

**Attention Word** – To help reject extraneous noises, each command to the robot should begin or end with “Eli”. If the robot ignores you the first time, just try the saying the command again or, alternatively, but the “Eli” attention word at the other end.

**Speaking Guidelines** – Try to be loud and enunciate clearly (like you were giving a lecture, or like an announcer on TV). Microsoft Speech likes there to be a silence interval at the end of each sentence so things like audience laughter can delay the robot’s response.

**Commands** – There are a variety of basic moves you can command the robot to perform, and a variety of ways you can specify your commands. You can try looking in the grammar file “sp\_drive.sgm” (a text file) to get a feel for the range of commands and types of phrases. Below is a list of standard commands, but many other variants are likely to work.

***Nod head*** – “Eli, nod your head.” / “Say yes, Eli.”

***Shake head*** – “Tossle your head, robot.” / “Shake your head please, Eli.”

***Look left*** – “Eli, look to the left.” / “Over here, bucket of bolts.”

***Raise arm*** – “Lift your arm, Eli.” / “Eli, go up now.”

***Extend arm*** – “Extend your hand, Eli.” / “Eli, reach out your arm.”

***Open hand*** – “Open your gripper, Eli.” / “Give it to me, bucket of bolts.”

***Grab something*** – “Hold this, Eli.” / “Eli, close your hand please.”

***Go forward*** – “Eli, advance please.” / “Step forward, robot.”

***Turn left*** – “Turn to your left, Eli” / “Rotate clockwise, Eli”

***Turn toward head and advance*** – “Come here, Eli.” / “Come on, robot.”

***Greeting*** – “Hello there, Eli” / “Eli, go to sleep now.”

***Say a canned phrase*** – “Eli, say something.” / “Speak now, bucket of bolts.”

***HAL-9000*** – “Open the pod bay doors, Eli.”

**Robot Responses** – For each particular conversational state, the robot randomly chooses between several responses. These are listed in the file “std\_quips.map” where “?0” is filled in by the user’s name. You can add or remove responses here by simply altering the file.

**Log File** – For each run a log file of the interaction is saved in the subdirectory “log”.

## Thursday

**Following Algorithm** – The robot uses its Kinect depth sensor to find the target person and follow him. It operates by creating a slice through space at roughly waist level, then looks for a distinct region that is roughly the right width for a human. In the GUI program you can see a red stripe draw across the region it thinks it should follow.

**Command Line** – The command line program “sp\_follow” just starts up the following routine. It takes an optional argument of which voice model to use (e.g. Jon-Connell-VT2). No images are displayed in this version so it is better for VNC operation. Note that it takes a while (30 seconds) to initialize all the robot hardware.

**GUI Version** – This is a development tool that displays intermediate results and allows for adjustment of many parameters. Double click the icon to start the application – it takes a while for the window to open during which the robot will twitch a bit. After this use the last option in the last menu (Demo / Speech Follow) to start the routine. Again, it will take a while to initialize the robot hardware, but the robot will announce when it is fully booted. The second text window with a similar name shows speech results during operation of the program.

**Speech Commands** – You can basically tell the robot to start following you or temporarily stop following you. As with the other demos, the robot needs to hear its name at the beginning or end of the command. You can look at the grammar file “gram\_follow.sgm” (text) to get a sense of what kind of variations are allowed. As a guide, some suitable sample phrases are given below:

***Start Following*** – “Eli, follow me please.” / “Over here, Eli.” / “Come along, Eli.”

***Stop Following*** – “Eli, stay right there.” / “Park it, Eli.” / “Eli, stop now please.”

***Turn Right*** – “On your right, Eli.” / “I’m on the right, Eli.”

***Turn Left*** – “Turn to the left, Eli.” / “Eli, rotate counterclockwise please.”

***Turn Around*** – “Eli, turn around.” / “Behind you, Eli.”

**Audio Localization** – If you are out of sight, the robot can use his directional microphone to turn towards you. This really works best within +/- 60 degrees, but you can call to it several times to make it turn further. Note that this function may be unpredictable if you are using a Bluetooth headset in a noisy environment (the robot may latch on to a whirring fan instead).

**Verbal Responses** – Generally the robot will confirm a change in state from following to not following, and vice versa. It will also complain if you walk too fast or it loses sight of you. There are a variety of wry comments listed for each such possible robot state (e.g. asked to follow but already in following mode). These are listed in file “std\_phrases.map” and can be edited by hand (the user’s name is substituted for “?0”). When there are several choices the robot picks randomly between them.

**Non-Responsiveness** – This program seems a little buggy (ver 2.30) and sometimes stops listening to the human. In this case click on the GUI window to generate a dialog box asking if you want to stop. Click “yes” then start Demo / Speech Follow again from the menus.

**Log File** – For each run a log file of the interaction is saved in the subdirectory “log”.

## Klepto

**Selecting a Surface** – The robot currently only uses a color 2D camera image to find objects. For this to work properly the robot needs a largely clear surface with a uniform color (not clear glass). The objects should be placed near the middle of this area about 12” in front of the arm, and clearly separated from each other.

**Objects for Manipulation** – The robot characterizes object by color distribution and overall size. We typically use empty OTC medication bottles like Advil, but add a few pennies inside to keep them from skittering across the table from a casual brush. Cooking spices would also work as long as they are less than about 8 ounces in weight. Make sure that the containers are fairly different in terms of colors since the robot cannot read labels (yet).

**Command Line** – This runs the basic speech interaction demo without showing any images (i.e. for VNC use). The executable takes as an optional argument some user voice model file name.

**GUI Version** – Double click to bring up the main window, then select the last option on the last menu (Demo / Speech Grab) to start the system. It takes a while to reset the robot and initialize the Kinect camera. After it is finished doing this the robot will announce that it is ready.

**Basic Commands** – You can designate things by size, position, or color and ask about these same properties plus counts. Examples include: “What color is the biggest object, Eli?” / “How many red things do you see, Eli?” / “Eli, pickup the thing in the middle, please.” / “What color is this <point> thing, Eli?” / “Where is the blue object, Eli?”

**Pointing** – Interaction is multimodal so you can designate a particular object by pointing to it. This works best if you actually touch the top of the object. The touching should be completed before you finish speaking the command (e.g. “Grab this <touch> object, Eli.”).

**Handoff Location** – When you ask the robot to give you something, it always transports it to a fixed location to the left of the arm. It then waits until it sees your hand to release the object. Note that it is expecting to have the object handed back to it later, and may become confused if this does not happen. Handoff phrases are like : “Give me it, Eli” / “Eli, I want the Tylenol.”

**Learning Nouns** – You can indicate an object by size, color, position, pointing, or pronoun. Generally you say: “Eli, the thing on the left is screen cleaner.” / “This <point> is called a soda can, Eli.” The name is then associated with a simple visual model and can be used in other commands like: “Where is the screen cleaner, Eli” / “Eli, what color is the Tylenol?” The system is set up to forget whatever it just learned when you exit the program.

**Learning Verbs** – The only action guaranteed to work is “poke” although “tap” might be possible. You start by saying something like: “Let me teach you how to poke something, Eli”. You then command individual actions like: “Point at it, Eli” then “Raise your hand, Eli”. Finally you finish the macro by saying: “That is how you do it, Eli”. The system is set up to forget whatever it just learned when you exit the program.

**Action Overrides** – This is the part in the video where the robot refuses to hand over aspirin, or offers Tums as a substitute for Roloids. This function used a socket connection to a server in Tokyo Research Lab that is no longer operational, thus these features are NOT available. A localized version of this backend could be constructed fairly easily if it is deemed an important demo functionality.



## Troubleshooting

**Audio Difficulties** – Sometimes the robot does not speak or hear properly because the audio settings on the laptop have gotten changed. Use Start / Control Panel to get to the “Sound” settings. In the “Playback” tab the default should be “Andrea PureAudio USB-SA” (speakers on the robot’s shoulders). In the “Recording” tab the default device should be “VoiceTracker II”.

**Poor Sensitivity** – If the robot only responds to speech if you are very close or shouting, check that the switch on the back of the silver microphone bar is set to “Line” (not “Mic”). To get this to take effect you may need to powercycle the robot or unplug then replug the USB cable from the back of the microphone.

**Mouth Barely Flashes** – Check that the laptop volume is at least halfway up. The volume of the robot’s speakers can be adjusted afterward using the volume control on the left speaker (black knob on inner surface).

**Subsystems Inoperable** – When moving the robot and control programs to a new computer, very often the serial ports will be assigned the wrong numbers. You can check this using “My Computer” then right click to select “Manage” and finally open “Device Manager” on the left. Look for the assignments under “Ports (COM & LPT)” and check COM5 (arm), COM6 (base), COM7 (TTL = mic), and COM8 (command = lift).

**Dead Servos** – Occasionally one of the servos in the arm (or neck) will overheat and stop functioning correctly. This is likely due to some sort of overload such as being rammed into a wall or table. Generally the servo will have a persistent red light on when this happens. These are Robotis AX-12A units and cost about \$50.

**Blown Motor Driver** – The other thing that sometimes breaks is the motor driver for the main wheels (e.g. if the robot slams into a wall full speed). Typically there is smoke and the robot starts spinning. This piece is best replaced with a 30A RoboClaw controller from Basic Micro (about \$130).

**Program Won’t Run** – Try completely rebooting the robot (described earlier) and trying again. Sometimes there is a problem talking to the arm & neck servo control device, especially if the previous run of the program crashed. The Kinect sensor also likes to be cold-booted when this happens.

**Reinstalling Software** – There is a directory c:\robot\install that contains all the necessary drivers if you need to move to a different laptop. The file “ELI\_ReadMe.txt” contains step-by-step instructions of how to do this.

